

Exercise 1 (5 points) :

1. Given the language L over the alphabet $\Sigma = \{0, 1\}$ which contains palindromes having a length less than 10,000. Is L regular ? justify your answer. (1 point)
2. Could we find an unambiguous grammar (CFG) for each context-free language? Justify your answer. (1 point)
3. Can we conclude that a language L is regular if a selected word "s" with a length greater than the pumping length "p" satisfies the pumping lemma without any contradictions? Justify your answer. (1 point)
4. Give an example of two context-free languages to show that the intersection is not closed under the set of context-free languages. (1 point)
5. In Turing machines, explain how the stay-put (S) transition $(q_0, 0) \Rightarrow (q_1, 1, S)$ can be implemented with only the conventional transitions {Left (L) , Right (R)} after each action. The tape alphabets are $\Gamma = \{0, 1, \square\}$ (where $\square$ is the blank symbol). (1 point)

Exercise 2 (5 points) :

1. Draw a DFA for the language that contains at least two ones. (1 point)
2. Let $D = \{w \mid w \text{ contains an even number of } 0\text{'s and an odd number of } 1\text{'s and does not contain the substring } 01\}$. Give the corresponding DFA with five states. (2 points)
3. Let $\Sigma = \{0, 1, +, =\}$ and $ADD = \{x=y+z \mid x, y, z \text{ are binary integers, and } x \text{ is the sum of } y \text{ and } z\}$. Show that ADD is **not** regular. (2 points)
Examples of words from the language ADD is $\{11=10+01, 100=11+1, \dots \text{etc} \}$

Exercise 3 (5 points) :

Let $\Sigma = \{0, 1\}$ and let B be the collection of strings that contain at least one 1 in their second half. In other words, $B = \{uv \mid u \in \Sigma^*, v \in \Sigma^*1\Sigma^* \text{ and } |u| \geq |v|\}$.

- a. Give a PDA that recognizes B . (2 points)
- b. Give a deterministic Turing machine that decides B . (3 points)

Exercise 4 (5 points) :

Given the Turing machine M which converts a binary representation of a number to its equivalent unary representation, and removes all symbols other than the unary representation from the tape.

Examples: $0 \Rightarrow \square, 1 \Rightarrow 1, 10 \Rightarrow 11, 101 \Rightarrow 11111$.

1. Write the algorithm for the Turing machine M in clear and precise English (Pseudo-code is accepted). (2 points)
2. Draw the state diagram for the Turing machine M . (2 points)
3. Run a simulation of the Turing machine M on input "10" showing the configuration table with each step as a show highlighting the current step and the position of the tape head. (1 point)